

FAHAD FAYAZ

Aspiring AI Engineer

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SUMMARY

Aspiring AI Engineer with hands-on experience building **LLM-powered applications**, **RAG pipelines**, and **deep learning** systems. Skilled in **Python**, **FastAPI**, **LangGraph**, and **FAISS**, with practical experience in embeddings, vector search, and prompt engineering. Strong foundation in computer vision and deep learning using **PyTorch**, with proven ability to package and deploy AI solutions as production-grade services and live applications.

SKILLS

Generative AI & LLMs: LLM Application Development, Retrieval Augmented Generation (RAG), Embeddings, Vector Search, Prompt Engineering, LangChain, LangGraph, FAISS, Hugging Face, OpenAI API

Machine Learning: Deep Learning, Computer Vision, NLP, PyTorch, Scikit-learn, OpenCV, Model Training & Evaluation

Programming & Engineering: Python (OOP, API development, debugging), SQL, FastAPI, Streamlit, REST APIs

Data & Big Data: Pandas, NumPy, PySpark, Databricks, MySQL

Cloud & DevOps: AWS EC2, AWS S3, Docker, Git, GitHub, CI/CD Fundamentals

EXPERIENCE

Sabudh Foundation

Jan 2026 – Jun 2026

Data Science Intern – Visual Search & Retrieval and Change Detection for Satellite Imagery (Remote)

- Developed a UCDNet deep learning model for 512×512×13 multi-band Sentinel-2 imagery, achieving **99.12% accuracy**, **87.62% F1 score**, 93.94% recall, and 79.47% Jaccard Index across 24 cities, with band-wise visualizations tailored to detect vegetation loss, avalanches, and river encroachment.
- Built a visual search and retrieval system extracting deep feature embeddings from satellite patches for content-based retrieval of spectrally similar regions across large image collections.
- Engineered the complete AI pipeline in PyTorch from multi-temporal data preprocessing through model training, evaluation, and deployment.

Purple Rocket

Jul 2025 – Dec 2025

Data Science Intern (Srinagar, J&K – Hybrid)

- Packaged and deployed trained image classification models into production-grade REST APIs using FastAPI for real-time inference.
- Built end-to-end ML pipelines covering data preprocessing, model training, and performance evaluation.
- Conducted exploratory data analysis on structured datasets to surface actionable business insights.

PROJECTS

PolicyBot – Enterprise Knowledge Assistant

2026

Python, LangGraph, FAISS, Streamlit, Hugging Face | Demo | GitHub

- Built a RAG-based Q&A chatbot that answers natural-language questions over uploaded company documents (grounded in source text), with persistent multi-session chat history in SQLite.
- Engineered a LangGraph retrieve-generate pipeline with FAISS vector search and sentence-transformer embeddings, served by the open-source Qwen 2.5 LLM via the Hugging Face Inference API, deployed through a live Streamlit interface.

UCDNet – Satellite Change Detection App

2026

PyTorch, FastAPI, Streamlit | Demo | GitHub

- Built and deployed a deep learning app for automated change detection in bi-temporal Sentinel-2 imagery using a UCDNet Siamese encoder-decoder, achieving **98.60% accuracy** and **82.37% F1 score**.
- Served the model via a FastAPI REST API and an interactive Streamlit demo with 53 pre-loaded satellite patches for live inference.

EDUCATION

Islamic University of Science and Technology, J&K

2020 – 2024

B.Tech in Computer Science and Engineering

Relevant Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, DBMS, Computer Networks, Operating Systems, Cloud Computing

CERTIFICATIONS

- Dataiku Core Designer, ML Practitioner, MLOps & Generative AI Practitioner Certificates** – Dataiku Academy
- HP LIFE:** Data Science & Analytics Certification
- Learn Python Programming – Beginner to Master** – Udemy